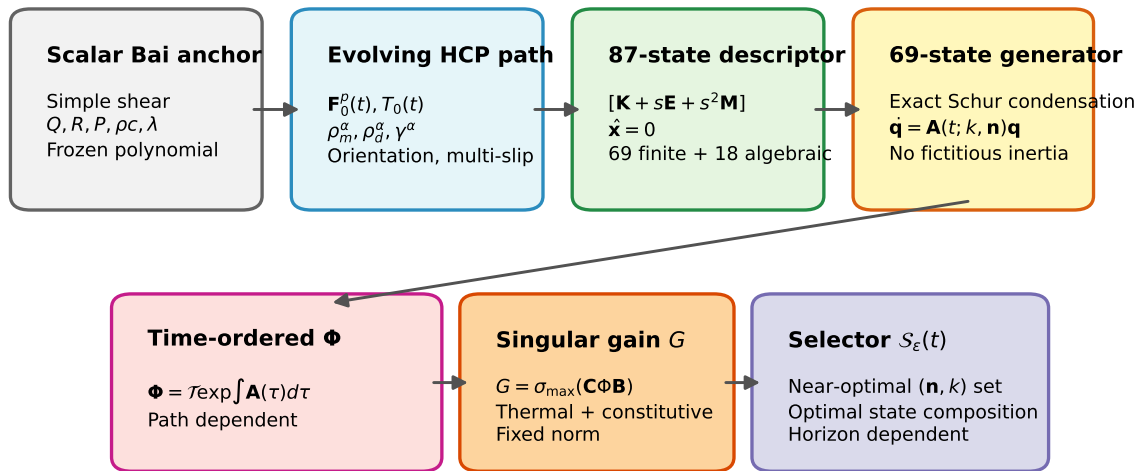




Classical theory cannot determine

which crystal perturbation accumulates the largest gain, or why direction, wavelength, and state composition change with the observation horizon.

Finite-time crystal theory resolves

maximum history-conditioned gain and a tested near-optimal direction--wavenumber--state set, subject to explicit norm, search-box, and validation boundaries.